

DAT32-2

Exploratory Data Analysis & Data Quality

Scoping, profiling, exploring, and reporting on data

Every data project begins with honest assessment of the question and the data, not with model fitting.

Albert School — 16 hours

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1 From business situation to data question

A data project begins not with data but with a vague business situation. The first task is to turn that situation into a specific, answerable analytical question.

Definition 1 Scoping a business problem means translating a vague situation into a **data question**: a question that names the quantity to be measured or relationship to be investigated, the population it concerns, and the data required to answer it.

Scoping requires deciding three things explicitly:

- **What data is needed** to answer the question, and whether it exists and is accessible.
- **What “good enough” looks like** — the level of accuracy, coverage, or confidence that would let the stakeholder act.
- **What is out of scope** — the questions the analysis will not attempt to answer.

A well-scoped question is specific enough that two analysts would agree on whether a given result answers it.

2 Systematic exploratory data analysis

Exploratory data analysis (EDA) is the systematic examination of a dataset before any modelling, proceeding from one variable to many.

Definition 2 **Univariate analysis** examines one variable at a time: its distribution, central tendency, spread, and shape.

Bivariate analysis examines the relationship between two variables.

Multivariate analysis examines relationships among three or more variables simultaneously.

Definition 3 **Anomaly detection** is the identification of observations that depart markedly from the rest of the data — values, records, or patterns that do not fit the distribution the rest of the data follows.

A systematic EDA proceeds in this order: profile each variable univariately, examine pairwise relationships, then consider multivariate structure, flagging anomalies at each stage.

3 Choosing visualizations

The chart must match the question being asked of the data.

Definition 4 The four EDA question types and their standard visualizations:

- **Distributions** (the shape of one variable): histogram, box plot, density plot.
- **Relationships** (how two variables move together): scatter plot, line of best fit.
- **Compositions** (how a whole divides into parts): bar chart, stacked bar chart, pie chart.
- **Time trends** (how a variable changes over time): line chart.

The choice of chart is justified by the EDA stage and the question: a distribution question is not answered by a chart designed to show composition.

4 Missing data mechanisms

Missing values are not all missing for the same reason, and the reason determines whether they may be imputed or must be excluded.

Definition 5 MCAR (Missing Completely At Random): the probability that a value is missing is unrelated to any data, observed or unobserved.

MAR (Missing At Random): the probability of missingness depends only on **observed** variables, not on the missing value itself.

MNAR (Missing Not At Random): the probability of missingness depends on the **unobserved** missing value itself.

The mechanism justifies the strategy. Under MCAR, dropping the affected rows does not bias the analysis. Under MAR, imputation conditioned on the observed variables is justified. Under MNAR, both dropping and naive imputation bias the result, and the missingness must itself be modelled or disclosed as a limitation.

5 Outlier handling

An extreme value is not automatically an error. Handling an outlier requires first deciding which of three kinds it is.

Definition 6 An outlier may be:

- a **data entry error** (a wrong value, e.g. a misplaced decimal) — to be corrected or removed;
- a **legitimate extreme** (a real but unusual observation) — to be kept;
- **distribution behaviour** (an expected tail value of a skewed or heavy-tailed distribution) — to be kept, with the distribution's shape acknowledged.

The handling decision follows from the classification: only errors are removed without further thought; legitimate extremes and tail values are retained, and removing them would distort the conclusions.

6 Correlation analysis

Definition 7 The **correlation** between two numeric variables measures the strength and direction of their linear association, on a scale from -1 to $+1$. A **correlation matrix** is the table of pairwise correlations among a set of variables.

A correlation matrix is read by scanning for entries near $+1$ or -1 , which indicate strong linear association.

Common pitfall Correlation does not imply causation. Two variables can be strongly correlated because one causes the other, because both are driven by a third variable, or by coincidence. A concrete example: ice-cream sales and drowning deaths are positively correlated, but neither causes the other — both rise with hot weather.

7 Feature engineering basics

Feature engineering creates new variables from existing ones to make structure in the data more usable.

Definition 8

- **Binning:** converting a continuous variable into discrete intervals (categories).
- **Log transformation:** replacing a variable by its logarithm, to compress a skewed range and linearise multiplicative relationships.
- **Interaction terms:** a new feature formed from the product (or combination) of two features, capturing an effect that depends on both.
- **Date feature extraction:** deriving features such as day of week, month, quarter, or is-weekend from a date or timestamp.

8 Data quality reporting

Profiling a dataset end-to-end produces a **data quality report** organised around four dimensions.

Definition 9 The four data quality dimensions:

- **Completeness:** the extent to which required values are present (the inverse of missingness).
- **Consistency:** the absence of contradictions within or across records (e.g. the same entity recorded the same way; values that agree with each other).
- **Uniqueness:** the absence of unintended duplicate records.
- **Timeliness:** whether the data is current enough for the question it is meant to answer.

A data quality report assesses each dimension over the dataset and states, for each, the issues found and their impact on the analysis.

9 Hypothesis formulation

Definition 10 A **data-driven hypothesis** is a specific, testable statement about the data suggested by the EDA — a relationship or pattern the exploration revealed, stated precisely enough to be checked against data.

Each hypothesis is documented together with a proposed **analytical approach** to test it: the data, comparison, or method that would confirm or refute it. The deliverable of an EDA includes at least five such documented hypotheses, each with its proposed approach.

10 Simpson's paradox

Definition 11 **Simpson's paradox** occurs when a trend that appears in aggregated data reverses or disappears when the data is broken down into subgroups.

The paradox arises when a third variable (a confounder) is distributed unevenly across the subgroups. It is the reason aggregated data can produce misleading conclusions: a relationship measured on the whole can be the opposite of the relationship that holds within every subgroup. Disaggregating by relevant subgroups is the check against it.

11 Analytical communication

The final deliverable is a written report that a non-technical stakeholder can act on.

Definition 12 An analytical report communicates findings through:

- a **narrative arc**: the question, what was found, and what it means, in an order a non-technical reader can follow;
- **visualization selection**: charts chosen to support each finding, matched to the question as in the EDA;
- **limitations disclosure**: an honest statement of what the data and analysis cannot support;
- **recommendations**: the actions the findings support.

The report is judged by whether the intended audience can act on it without needing the underlying code or statistics.

Exercises

1. Take a vague business situation and scope it into a specific, answerable data question. State what data is needed, what success looks like, and what is out of scope.
2. Design and execute a systematic EDA on a dataset: univariate, bivariate, and multivariate analysis, and anomaly detection.
3. For each EDA stage, choose an appropriate visualization (distribution, relationship, composition, time trend) and justify the choice.
4. For a dataset with missing values, characterize the missingness as MCAR, MAR, or MNAR, and justify an imputation or exclusion strategy.
5. Detect outliers in a variable and classify each as a data entry error, a legitimate extreme, or distribution behaviour, then decide how to handle it.

- 6.** Compute a correlation matrix and interpret it. Give a concrete example of two correlated variables that are not causally related.
- 7.** Apply binning, log transformation, an interaction term, and date feature extraction to a dataset.
- 8.** Profile a dataset end-to-end and write a data quality report covering completeness, consistency, uniqueness, and timeliness.
- 9.** From an EDA, formulate and document at least five data-driven hypotheses, each with a proposed analytical approach to test it.
- 10.** Construct or identify a case of Simpson's paradox and explain why the aggregated conclusion misleads.
- 11.** Write an analytical report for a non-technical audience communicating findings, visualizations, limitations, and recommendations.